Free Insurance, Fewer Claims: Strategic Enrollment in Nepal's Health Insurance

Nepal made health insurance free for citizens over 70. Surprisingly, their healthcare claims *fell*. Using administrative data on 40 million claims from 10 million enrollees and a regression discontinuity (RD) design, I show that annual claims per enrollee drop by 0.86, a 33% reduction, at the eligibility threshold (Figure 1). Since the data are conditional on enrollment, this discontinuity characterises selection into insurance—how the *composition* of the insured pool changes at the threshold—rather than insurance effects on utilisation. Individuals with higher healthcare needs enroll before age 70 by paying premiums to access benefits immediately, whereas individuals with lower healthcare needs enroll when it is free.

The insured population just above 70 is therefore systematically healthier than those just below. An ex-ante expected cost index, estimated on re-enrollees aged 50–65 (away from the threshold) using their pre-analysis-period claims history including diagnoses, hospitalisation, and care intensity, confirms this composition change (Figure 2).

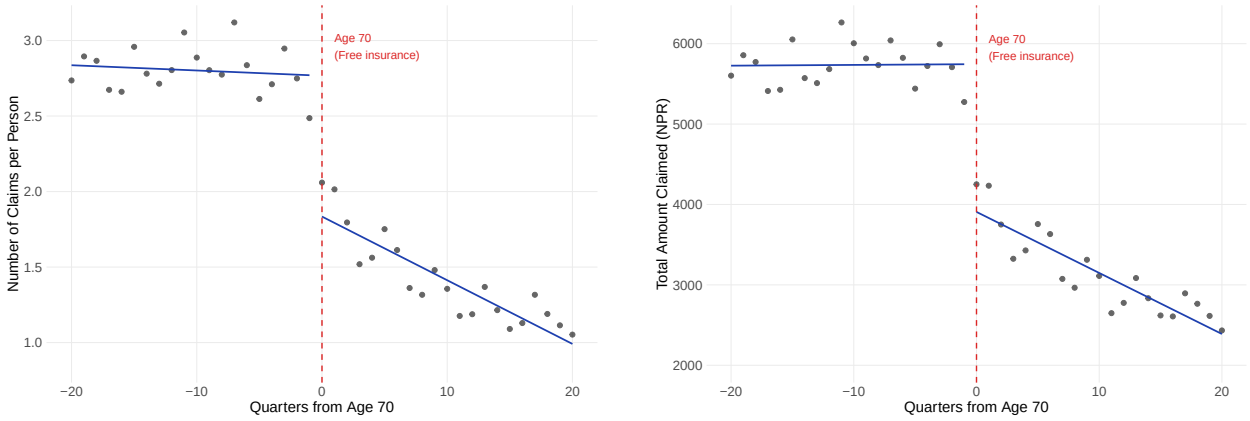
Why This Matters. Universal Health Coverage remains elusive even when insurance is fully subsidised (Banerjee et al. 2021; Levine, Polimeni, and Ramage 2016). Policymakers have proposed household bundling and restricted enrollment windows as solutions to adverse selection (Fischer, Frölich, and Landmann 2023). Nepal's National Health Insurance Program (NHIP) implements both enrollment at the household level, and restricted enrollment windows. Yet strategic behaviour persists. These results challenge the prevailing policy recommendation in low-income health insurance markets: even with household bundling and restricted enrollment windows, substantial adverse selection persists at predictable thresholds. The core problem is not selection within an enrollment period, but selection *across* predictable eligibility thresholds.

Separating Selection from Moral Hazard. Separating selection from moral hazard is a central identification challenge (Chiappori and Salanié 2000; Einav and Finkelstein 2011). Age-based eligibility thresholds provide a natural setting for this test (Card, Dobkin, and Maestas 2008). I address this challenge by combining two features. First, Nepal's age-70 threshold, introduced in April 2019, creates sharp price variation. Individuals just below pay full premiums, and those just above receive free insurance. Second, I construct an expected cost index from claims data preceding

the analysis period (before May 2023), predicting utilisation from diagnoses, hospitalisation history, and care intensity. Because the index is constructed using claims data before my main analysis period (May 2023–May 2025), any discontinuity at age 70 reflects selection into coverage rather than behavioural responses to it. A placebo test using 2016–2018 data, before the age-70 subsidy was implemented, finds no discontinuity, confirming that the selection pattern is policy-induced.

Data. I use the universe of NHIP claims and registrations from 2016 through May 2025—over 40 million reimbursement claims from approximately 10 million individuals. The data include individual identifiers, exact birth dates, claim amounts, diagnosis codes, and hospital visits. My main analysis focuses on *re-enrollees* observed over their full insurance period between May 2023 and May 2025. Restricting to re-enrollees ensures that the expected cost index includes only individuals with a history of insurance coverage, avoiding underestimation of risk for new enrollees. This restriction also guarantees a common history of prior coverage, improving comparability around the threshold. Because the analysis conditions on enrollment, a mechanical increase in density at age 70 is expected. I therefore assess RD validity using population census data and predetermined characteristics, finding no discontinuities in age density, employment, or other observables at the threshold. The main estimation sample includes 570,904 re-enrollee observations using a data-driven bandwidth.

Figure 1 — Claims outcomes for re-enrollees



(a) Number of claims per enrollee (Annual) **(b) Total amount claimed per enrollee (Annual)**

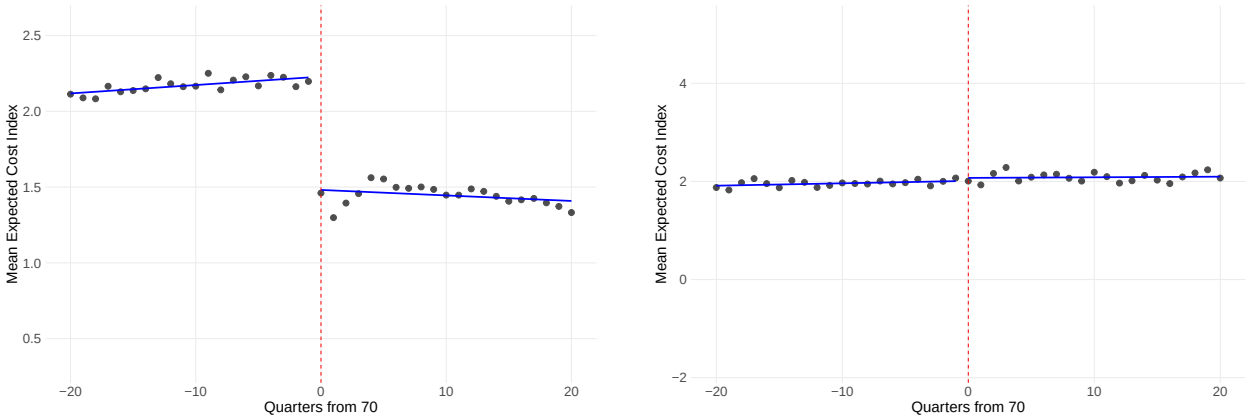
Note: Sample restricted to individuals who renewed their insurance. Each panel shows binned averages by age in quarters. The vertical dashed line indicates the age 70 cutoff.

Results. Figure 1 shows the main findings: annual claims per enrollee fall by 33% relative to the below-70 mean (RD estimate: -0.86, $p < 0.01$), and total claimed amount per enrollee declines by 31% relative to the below-70 mean of 5,173 Nepalese rupees (NPR) (RD estimate: NPR $-1{,}604$, $p <$

0.01). Consistent patterns emerge across other margins: the share making any claim drops by 12.2 percentage points and mean claim amount falls by 21%. These results are robust to alternative bandwidths, polynomial orders, weighting schemes, and inclusion of covariates.

Figure 2 reveals why. The ex-ante expected cost index drops discontinuously at age 70. Enrollees above the threshold are systematically healthier. High-risk individuals enroll before 70 to access benefits immediately; low-risk individuals wait for free coverage.

Figure 2 — Expected Cost Index for re-enrollees



(a) Expected Cost Index: 2024 **(b) Expected Cost Index: 2018**

Note: Expected cost index constructed from pre-analysis-period claims data; higher values indicate greater predicted healthcare utilisation. Each point represents the mean expected cost index of enrollees within a quarterly age bin. Sample restricted to re-enrollees. Panel (a) shows the expected cost index for 2024. Panel (b) uses a placebo expected cost index constructed from 2016–2018 pre-policy claims; no discontinuity appears at age 70.

Policy Implications. Threshold-based subsidies create new selection margins. When eligibility is predictable and enrollment is voluntary, individuals time their enrollment strategically—even with household bundling. This concentrates high-risk individuals among premium payers, raising costs; NHIP has already experienced payment delays linked to fiscal strain. The main lesson is that generous subsidies alone cannot achieve universal coverage without enforcement. Complementary policies—continuous coverage incentives, phased-in subsidies, or risk adjustment—are necessary to maintain balanced risk pools.

References

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